

Pleasant Grove
UT, 84062

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Isaac Crossman

Career Summary

Mechanical Designer with 4+ years of experience in the field of MEP design, drafting, and project coordination. Proficient in Autodesk Revit, Autodesk Fusion 360, HVAC system layout, understanding construction documents, 3D modeling, and coordinating with various disciplines. Well versed backgrounds with design and exposure in the field prioritizing efficiency, accuracy, and meeting deadlines.

Experience

Mechanical & Plumbing Designer & Bim Lead

2023 – Present Brunner & Dallon Engineering Sandy, UT

- Design and model HVAC and plumbing systems in Autodesk Revit
- Perform HVAC load calculations using CHVAC
- Size mechanical systems based on loads
- Coordinate with architects and other trades to tackle design conflicts
- Develop detailed construction documents
- Assist in equipment selection
- Route and size plumbing piping
- Fill out equipment and connection schedules
- Developed Revit templates to maximize consistency & efficiency throughout projects
- Develop and maintained Revit standards, families, and workflows
- Established modeling standards and workflows for projects
- Improve drafting efficiency through template optimization
- Support team members in Revit troubleshooting
- Participated in project coordination meetings
- Reviewed construction documents for quality control
- Hosted company wide Revit training to educate on standard workflows
- Developed strong communication and problem solving skills

Mechanical & Plumbing Drafter

2021 – 2023 Brunner & Dallon Engineering Sandy, UT

- Created and updated mechanical drawings using Revit and AutoCAD
- Assisted engineers with drafting HVAC layouts and design
- Modified construction documents for project revisions
- Learned industry standards for mechanical & plumbing design

Low Voltage Technician	2019 - 2021	AV Zone	Sandy, UT
	▪ Installed, configured, and troubleshooted low voltage systems including audio / video, network, and camera systems. ▪ Ran and terminated wiring for residential and commercial jobs ▪ Worked directly with clients to make sure the system was functioning as expected.		

Projects

Smart Interactive Cornhole Boards	2026	UVU Personal Project	Orem, UT
	▪ Designed and built custom cornhole boards integrating electronics and sensor systems ▪ Developed using ESP32 microcontroller for input processing and LED control ▪ Implemented break beam, vibration, and load cell sensors to trigger LED responses ▪ Designed and wired electrical system ▪ Troubleshooted hardware and code issues		

Relevant Skills & Technical Interests

- Electrical & Low Voltage Systems
- Microcontroller (Arduino / ESP32)
- Basic Circuit Design & Troubleshooting
- Autodesk Fusion 360
- Autodesk Revit
- Built and modified hobby-grade RC vehicles, including electrical & mechanical troubleshooting
- Experience with drone setup, operation, and maintenance
- Woodworking projects involving design, fabrication, and finishing
- 3D modeling
- 3D printing for prototyping custom parts

Education	Electrical Engineering (In Progress)	Utah Valley University	Orem, UT
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